Ruleout Threshold Robustness for the 1.1 Quarterly Event Study

BoardPharma

November 10, 2026

1 Ruleout Definition and Common Regression Design

For each saved cohort panel, observations are grouped by company and product (`BoardName` × `product`). Using all observed quarters in that cohort file, the ruleout score for company b, product p, and cohort c is

$$R_{bpc} = \frac{\max(\texttt{price1}_{bpc}) / \min(\texttt{price1}_{bpc})}{\max(\texttt{price0}_{bpc}) / \min(\texttt{price0}_{bpc})}.$$

For a threshold τ, every row of the company–product history is removed from that cohort when $R_{bpc} > \tau$. Equality is retained. The score uses the original price levels and is calculated before upper-p95 winsorization and the log transformation. Ruleout = 0 is the unfiltered benchmark. Ruleout values 50, 10, and 5 apply the same rule with progressively stricter cutoffs.

Treatment status, control status, event timing, event cohort, and dynamic ATC3 sharing are read from the same saved cohort files and are not reconstructed by ruleout. After removal, a cohort with no remaining treated observations is not stacked. The retained sample is then processed by the common regression code, including upper-p95 winsorization, logs, fixed effects, clustering, and `did_imputation` autosample selection.

All four runs use the same inputs and regression settings: the `drop_sentinel_yearquarter` roster, `ever_board=1`, `ieoc_ref=1`, `late=3`, new requirement, stay, and sharing-time rules, dynamic ATC3 sharing, `price1`, req2, the Not control group, company–product–cohort fixed effects, cohort–calendar-quarter fixed effects, company clustering, horizons 0–7, three pre-event coefficients, and no Kappa, other-event, or ATC3 fixed-effect controls.

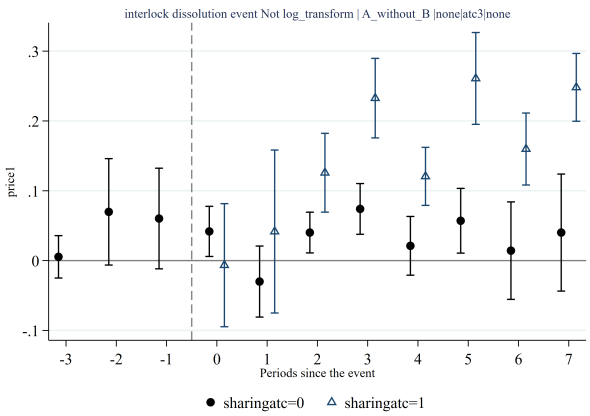
Table 1: Estimation sample across the six event–side regressions

Ruleout setting	Total observations	Reduction from ruleout = 0
0 (off)	583,584	0.00%
50	577,920	0.97%
10	568,884	2.52%
5	546,300	6.39%

Notes. Counts sum the final estimation observations reported by the six output files and therefore refer to stacked regression observations, not unique company–product records.

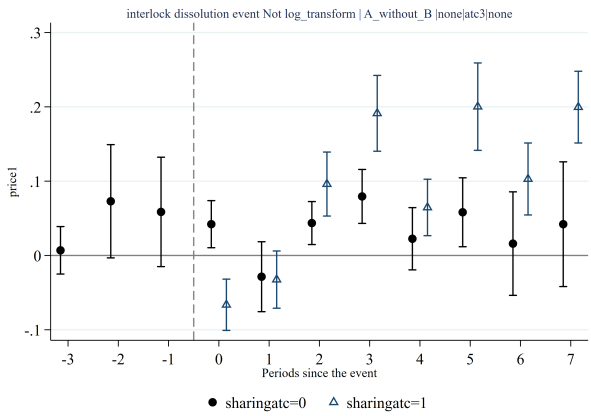
2 Interlock Dissolution — A-side Treatment

Ruleout = 0



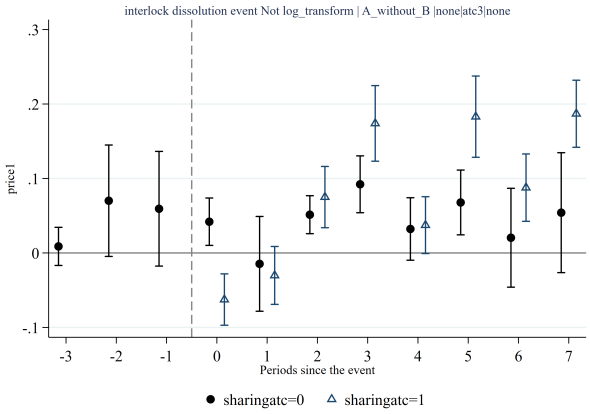
Estimation observations: 109,824

Ruleout = 50



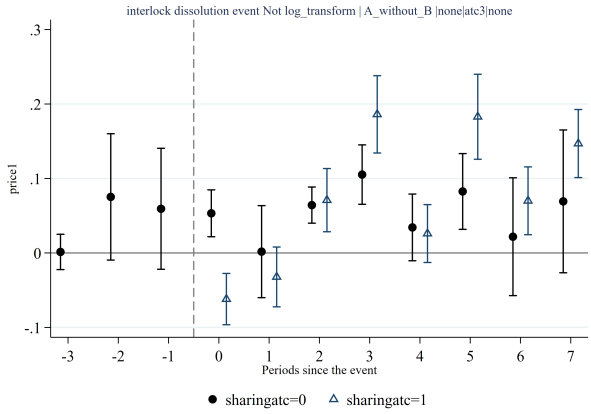
Estimation observations: 108,576

Ruleout = 10



Estimation observations: 106,932

Ruleout = 5

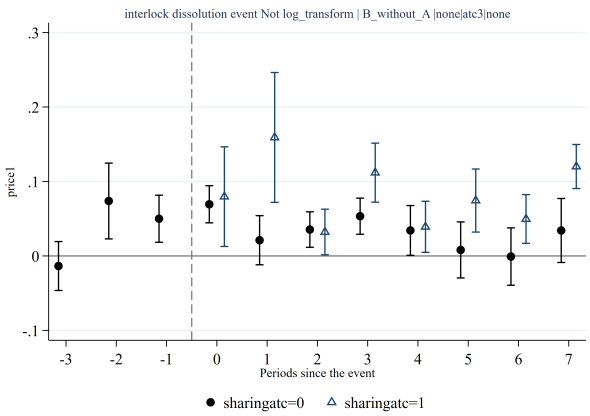


Estimation observations: 103,704

Notes. Black circles report estimates for `sharingatc=0`; hollow blue triangles report estimates for `sharingatc=1`. Error bars are 95% confidence intervals. The panels differ only in the ruleout threshold and its resulting estimation sample.

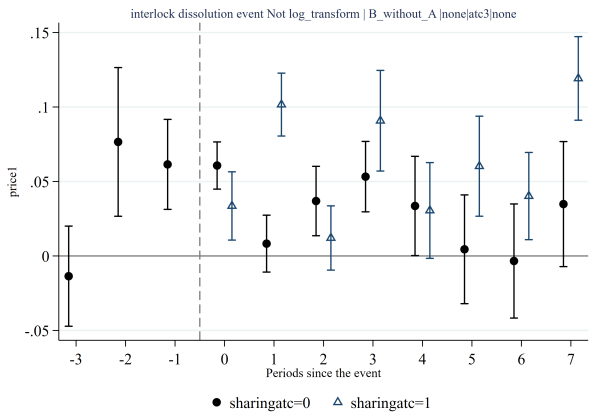
3 Interlock Dissolution — B-side Treatment

Ruleout = 0



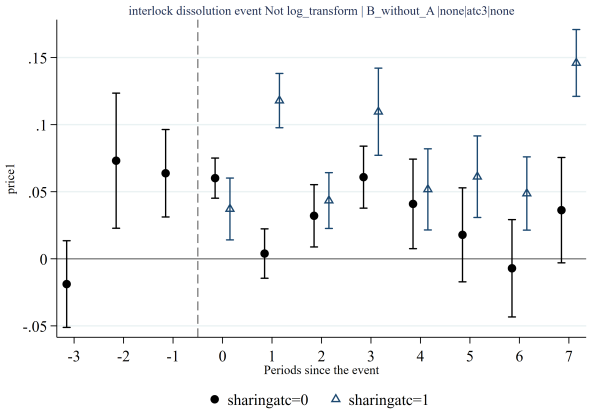
Estimation observations: 94,944

Ruleout = 50



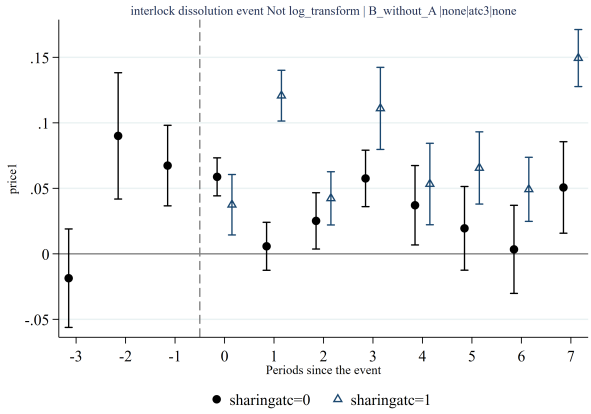
Estimation observations: 94,176

Ruleout = 10



Estimation observations: 92,760

Ruleout = 5

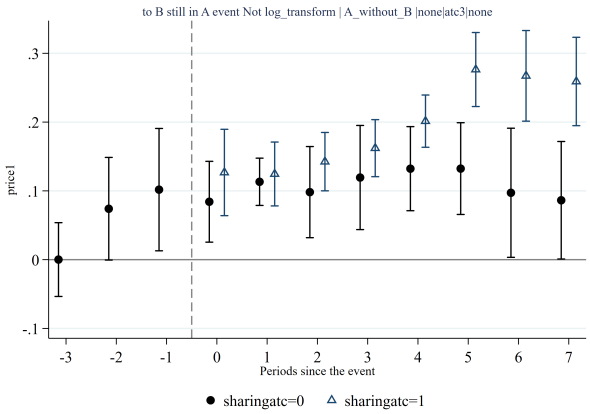


Estimation observations: 89,772

Notes. Black circles report estimates for `sharingatc=0`; hollow blue triangles report estimates for `sharingatc=1`. Error bars are 95% confidence intervals. The panels differ only in the ruleout threshold and its resulting estimation sample.

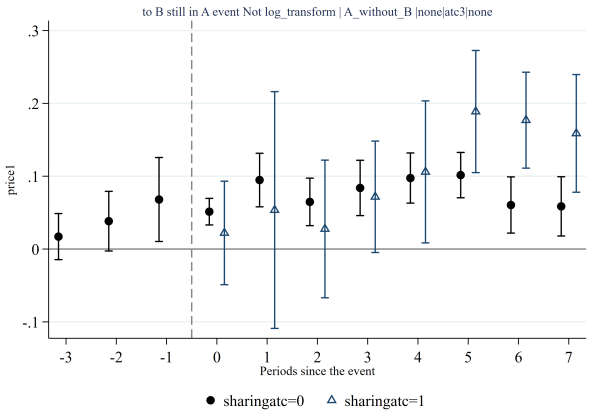
4 To B, Still in A — A-side Treatment

Ruleout = 0



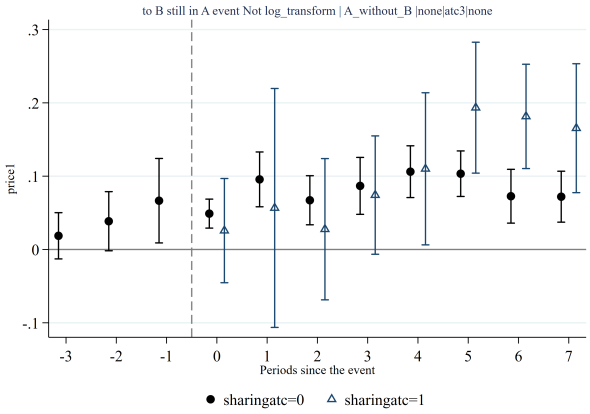
Estimation observations: 108,396

Ruleout = 50



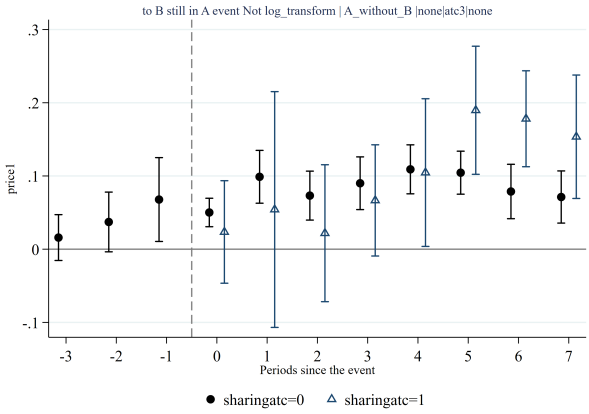
Estimation observations: 107,604

Ruleout = 10



Estimation observations: 105,864

Ruleout = 5

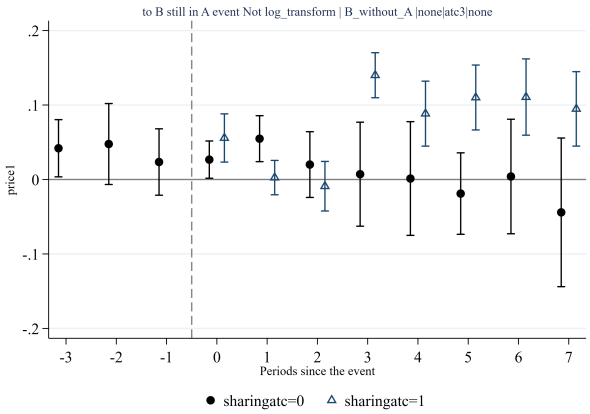


Estimation observations: 99,696

Notes. Black circles report estimates for `sharingatc=0`; hollow blue triangles report estimates for `sharingatc=1`. Error bars are 95% confidence intervals. The panels differ only in the ruleout threshold and its resulting estimation sample.

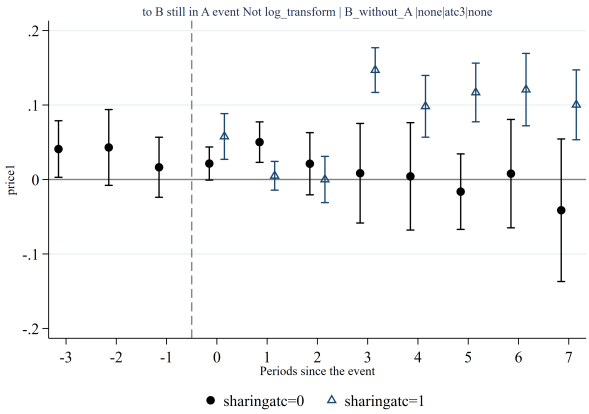
5 To B, Still in A — B-side Treatment

Ruleout = 0



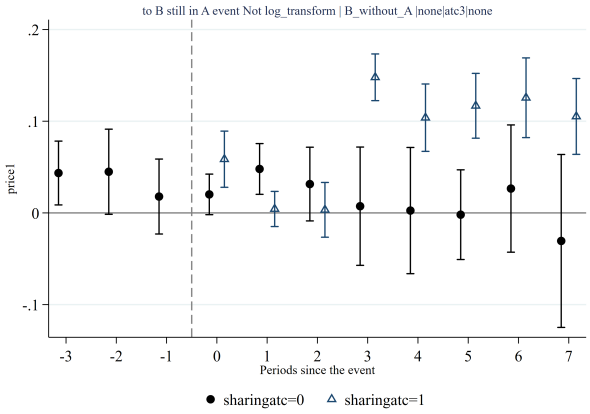
Estimation observations: 130,236

Ruleout = 50



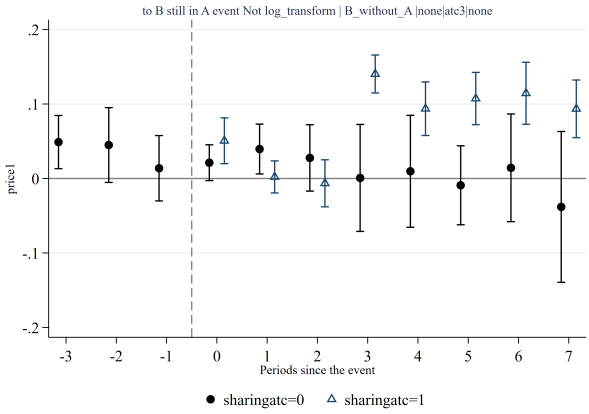
Estimation observations: 128,832

Ruleout = 10



Estimation observations: 126,840

Ruleout = 5

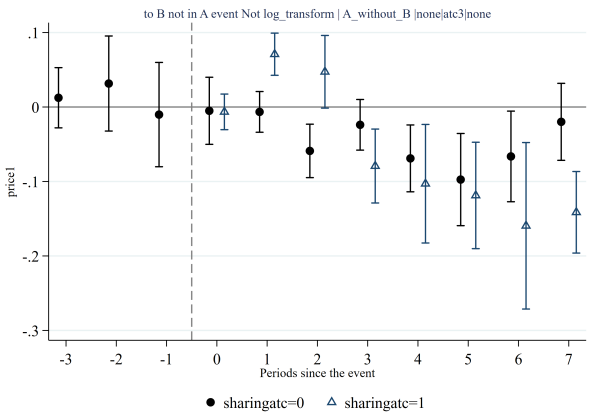


Estimation observations: 121,956

Notes. Black circles report estimates for `sharingatc=0`; hollow blue triangles report estimates for `sharingatc=1`. Error bars are 95% confidence intervals. The panels differ only in the ruleout threshold and its resulting estimation sample.

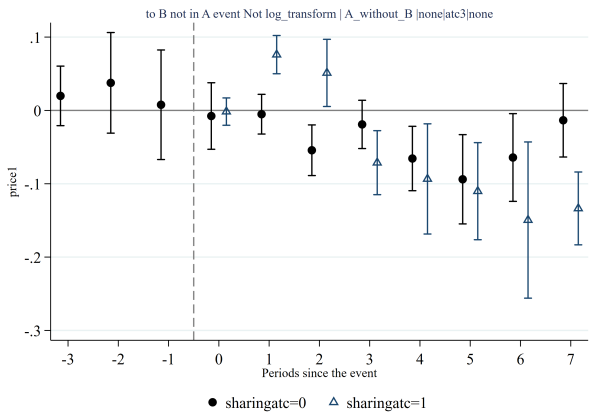
6 To B, Not in A — A-side Treatment

Ruleout = 0



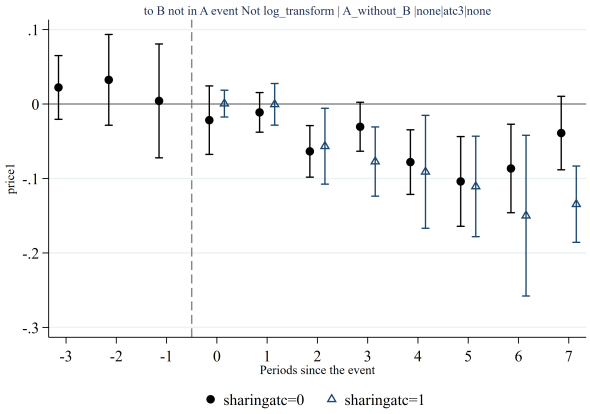
Estimation observations: 73,992

Ruleout = 50



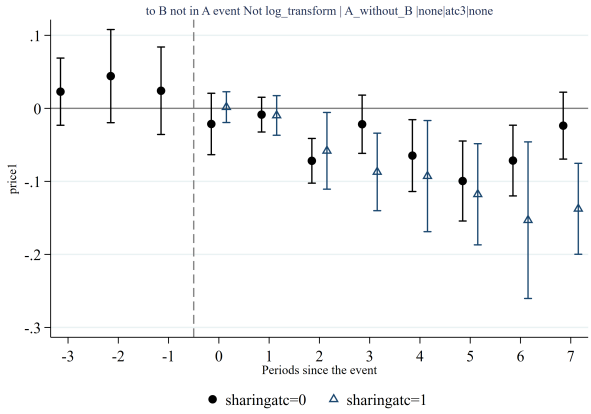
Estimation observations: 73,068

Ruleout = 10



Estimation observations: 72,048

Ruleout = 5

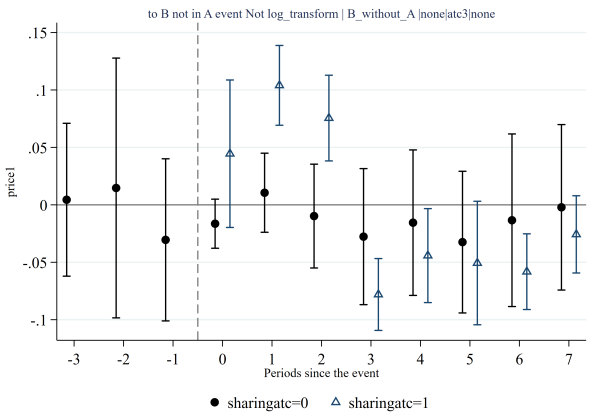


Estimation observations: 69,420

Notes. Black circles report estimates for `sharingatc=0`; hollow blue triangles report estimates for `sharingatc=1`. Error bars are 95% confidence intervals. The panels differ only in the ruleout threshold and its resulting estimation sample.

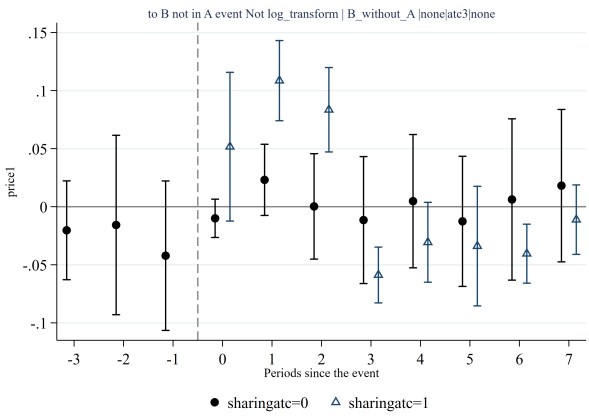
7 To B, Not in A — B-side Treatment

Ruleout = 0



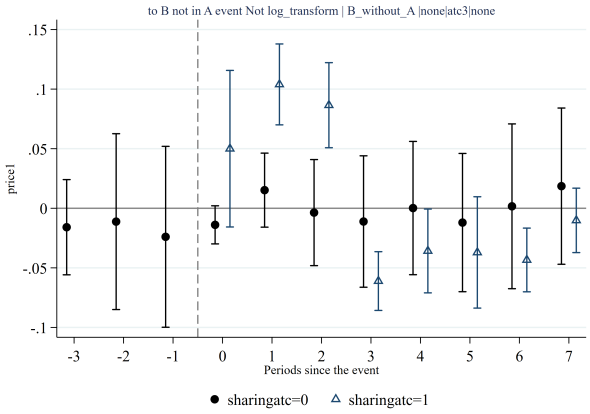
Estimation observations: 66,192

Ruleout = 50



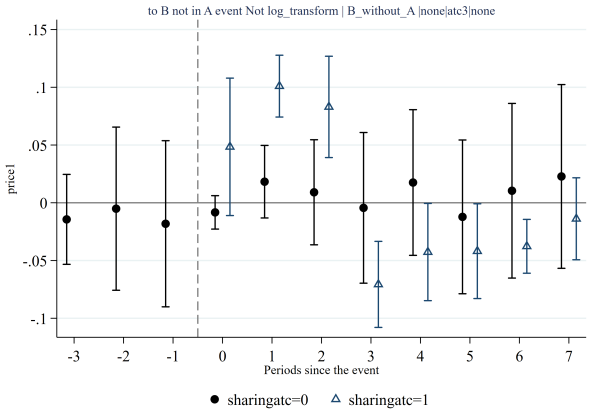
Estimation observations: 65,664

Ruleout = 10



Estimation observations: 64,440

Ruleout = 5



Estimation observations: 61,752

Notes. Black circles report estimates for `sharingatc=0`; hollow blue triangles report estimates for `sharingatc=1`. Error bars are 95% confidence intervals. The panels differ only in the ruleout threshold and its resulting estimation sample.